

0.1 Exercises: Sequences of Numbers (I)

0.1.1 Find the following limits:

$$(1) \lim_{n \rightarrow \infty} \frac{1}{n} \prod_{k=n!}^{(n+1)!} \sqrt[k]{e};$$

$$(2) \lim_{n \rightarrow \infty} \sum_{k=n^n}^{(n+1)^n} \frac{1}{k};$$

$$(3) \lim_{n \rightarrow \infty} \sum_{k=2^n}^{2^{(n+1)}} \frac{1}{k}.$$

0.1.2 Find the sets of limit points of the following sequences:

$$(1) \sin \ln n;$$

$$(2) \cos \sqrt{n};$$

$$(3) \tan\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right);$$

$$(4) \sqrt{n} - [\sqrt{n}].$$

0.1.3 Find the following limits:

$$(1) \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^a}{n^{a+1}}, \quad a > -1;$$

$$(2) \lim_{n \rightarrow \infty} \sum_{k=n+1}^{\infty} \frac{n^a}{k^{a+1}}, \quad a > 0;$$

$$(3) \lim_{n \rightarrow \infty} n \left(\frac{1}{a+1} - \sum_{k=1}^n \frac{k^a}{n^{a+1}} \right), \quad a > 0;$$

$$(4) \lim_{n \rightarrow \infty} n \left(\frac{1}{a} - \sum_{k=n+1}^{\infty} \frac{n^a}{k^{a+1}} \right), \quad a > 0;$$

0.1.4 Consider the sequence $f_0 : \mathbb{N}^* \rightarrow \mathbb{R}$,

$$f_0(n) = \left(1 + \frac{1}{n}\right)^n,$$

and define

$$f_k(n) = \left(\frac{f_{k-1}(n)}{\lim_{p \rightarrow \infty} f_{k-1}(p)} \right)^n, \quad n, k \in \mathbb{N}^*.$$

Find $\lim_{n \rightarrow \infty} f_k(n)$.

0.1.5 Find the following limits

$$(1) \lim_{n \rightarrow \infty} \sum_{k=1}^n \sin \frac{k}{n^2};$$

$$(2) \lim_{n \rightarrow \infty} \sum_{k=1}^n \sin^2 \frac{\sqrt{k}}{n};$$

$$(3) \lim_{n \rightarrow \infty} \sum_{k=1}^n (2^{k/n^2} - 1);$$

$$(4) \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{k}{n^2}\right).$$

0.1.6 Let (a_n) be a sequence such that

$$\lim_{n \rightarrow \infty} n(a_{n+1} - a_n) = K, \quad K > 0.$$

Prove that

$$\lim_{n \rightarrow \infty} a_n = \infty.$$

0.1.7 Let $f : [0, \infty) \rightarrow [0, \infty)$ be a function possessing a continuous derivative of order 2. Suppose that the following conditions are satisfied:

- (1) $f(x) \leq x, \forall x \geq 0$;
- (2) $f(0) = 1, f''(0) \neq 0$.

Define a sequence (a_n) by the recurrence formula

$$\frac{a_{n+1}}{n} = f\left(\frac{a_n}{n}\right), \quad n \geq 1, \quad a_1 \geq 0.$$

Prove that $\lim_{n \rightarrow \infty} a_n = 0$.

0.1.8 Find the limit of each of the sequences defined by the following recurrence formulas:

- (1) $\frac{a_{n+1}}{n} = \ln\left(1 + \frac{a_n}{n}\right), \quad n \geq 1, \quad a_1 \geq 0$;
- (2) $\frac{b_{n+1}}{n+1} = \ln\left(1 + \frac{b_n}{n}\right), \quad n \geq 1, \quad b_1 > 0$.

0.1.9 Find a sequence (a_n) of real number such that the sequence $(\sqrt[n]{a_n})$ is convergent but the sequence $\left(\frac{a_{n+1}}{a_n}\right)$ has not limit.

0.1.10 Let (a_n) be a sequence of positive real numbers such that

$$\limsup \frac{a_{n+1}}{a_n} < 1.$$

Prove that

$$\lim_{n \rightarrow \infty} a_n = 0.$$

0.1.11 Prove that, if $\sin x \neq 0$, then the sequence $(\sin nx)_{n \geq 0}$ has not limit.

0.1.12 Let (a_n) be a sequence of real numbers and $q \in (0, 1)$ such that

$$\lim_{n \rightarrow \infty} (a_{n+1} - qa_n) = 0.$$

Prove that $\lim_{n \rightarrow \infty} a_n = 0$.

0.1.13 Find the limit of the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^{1+\frac{1}{n}} \left(1 + \frac{1}{n+1}\right)^{1+\frac{1}{n+1}} \cdots \left(1 + \frac{1}{2n}\right)^{1+\frac{1}{2n}}.$$

0.1.14 Find the limit of each of the following sequences:

1. $x_n = n - n^2 \ln \frac{n+1}{n};$
2. $x_n = \left(n \left(\arctan \frac{\sqrt{n}}{\sqrt{n} + \sqrt{2}} - \arctan \frac{\sqrt{n} - \sqrt{2}}{\sqrt{n}} \right) \right)^{n^2};$
3. $x_n = \frac{(n+1)^a - n^a - an^{a-1}}{n^{a-2}}, \quad a \in \mathbb{R};$
4. $x_n = \sqrt[n]{n};$
5. $x_n = \frac{\sqrt[n]{n!}}{n};$
6. $x_n = \left(\frac{\ln(n+1)^{n+1}}{\ln n^n} \right)^n;$
7. $x_n = \frac{1}{n} \int_1^n \left(1 + \frac{1}{x} \right)^x dx;$
8. $x_n = n \sin(2\pi en!);$
9. $x_n = n^2 \left(\left(1 + \frac{1}{n} \right)^{n+0.5} - e \right);$
10. $x_n = \left({}^{n+1}\sqrt{(n+1)!} - \sqrt[n]{n!} \right);$
11. $x_n = \sum_{k=1}^n \frac{\sin k}{n+k};$
12. $x_n = n^{\ln \frac{n+a}{n+b}}, \quad a, b \in \mathbb{R};$
13. $x_n = n \ln \frac{a\sqrt{n+1} + b\sqrt{n+2} + c\sqrt{n+3}}{a\sqrt{n+2} + b\sqrt{n+3} + c\sqrt{n+4}}, \quad a, b, c \in \mathbb{R};$
14. $x_n = n^2 \left(a^{\frac{1}{n+b}} - a^{\frac{1}{n+c}} \right), \quad a > 0, \quad b, c \in \mathbb{R};$
15. $x_n = n^a \frac{\sqrt[n]{n} - 1}{\ln n}, \quad a \in \mathbb{R};$
16. $x_n = \sqrt[n]{\binom{n}{1} \binom{n}{2} \cdots \binom{n}{n-1}};$
17. $x_n = \frac{a^{6n}}{(1+a^n)(1+2a^{2n})(1+3a^{3n})}, \quad a > 0;$
18. $x_n = \frac{\ln n!}{n^a}, \quad a > 1;$
19. $x_n = \frac{[na]}{n}, \quad a \in \mathbb{R};$
20. $x_n = \frac{1}{\binom{n}{0}} + \frac{1}{\binom{n}{1}} + \cdots + \frac{1}{\binom{n}{n}};$
21. $x_n = \left(1 + \frac{a}{n} \right) \left(1 + \frac{2a}{n} \right) \cdots \left(1 + \frac{na}{n} \right), \quad a > 0;$

22. $x_n = \frac{1^r + 3^r + \cdots + (2n-1)^r}{n^{r+1}}, \quad r > 0;$
23. $x_n = \frac{1}{n} + \frac{1}{2n} + \cdots + \frac{1}{n^2};$
24. $x_n = \frac{1}{\ln 2^n} + \frac{1}{\ln 3^n} + \cdots + \frac{1}{\ln n^n};$
25. $x_n = \frac{1}{\ln n^2} + \frac{1}{\ln n^3} + \cdots + \frac{1}{\ln n^n};$
26. $x_n = \sum_{k=1}^n \left(\frac{1}{3k-2} + \frac{1}{3k-1} - \frac{2}{3k} \right);$
27. $x_n = \sum_{k=1}^n \frac{3k^2 + 3k + 1}{k^3(k+1)^3};$
28. $x_n = \sum_{k=1}^n \frac{k+2}{k(k+1)2^k};$
29. $x_n = \sum_{k=1}^n \frac{k!}{(k+p+1)!}, \quad p \in \mathbb{N}^*;$
30. $x_n = \sum_{k=1}^n \frac{k-2}{2^k};$
31. $x_n = \sum_{k=1}^n \frac{k 2^k}{(k+2)!};$
32. $x_n = \sum_{k=1}^n \frac{a^{k-1}}{(1+a^k)(1+a^{k+1})}, \quad a \in \mathbb{R} \setminus \{-1, 0, 1\};$
33. $x_n = \sum_{k=1}^n \frac{1}{\sqrt[k]{n^k + 1 + 1}};$
34. $x_n = \sum_{k=1}^n \frac{k^2}{k!};$
35. $x_n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n};$
36. $x_n = \frac{n^2+1}{n^2-1} \cdot \frac{n^2+2}{n^2-2} \cdots \frac{n^2+n}{n^2-n};$
37. $x_n = \frac{2a x_{n-1}}{a + x_{n-1}}, \quad a > 0, n \geq 1, x_0 > 0;$
38. $x_n = x_{n-1}^2 - 2x_{n-1} + 2, \quad x_0 \in \mathbb{R}, n \geq 1;$
39. $x_n = e^{-1+x_{n-1}}, \quad x_0 \in \mathbb{R}, n \geq 1;$
40. $x_n = 2^{\frac{x_{n-1}}{2}}, \quad x_0 \in \mathbb{R}, n \geq 1;$
41. $x_n = e^{\frac{x_{n-1}}{e}}, \quad x_0 \in \mathbb{R}, n \geq 1;$
42. $x_n = \frac{1}{2} \left(x_{n-1} + \frac{a}{x_{n-1}} \right), \quad x_0 > 0, a > 0, n \in \mathbb{N};$

43. $x_n = \sqrt{x_{n-1}x_{n-2}}, \quad x_0 = x_1 = 1, \quad n \geq 2;$

44. $x_n = a^{1+\ln x_{n-1}}, \quad x_0 = a > 0, \quad n \geq 1;$

45. $x_n = n \left(\left(1 + \frac{a}{n}\right)^n - \left(1 + \frac{1}{n}\right)^{an} \right), \quad a \in \mathbb{R};$

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46. $x_n = n \left(\frac{1}{e-1} - \frac{1^n + 2^n + \dots + (n-1)^n}{n^n} \right);$

0.1.15 Let $(x_n)_{n \geq 1}$ be a sequence. Prove that the sequence $(y_n)_{n \geq 1}$,

$$y_n = \frac{1}{n} \ln (e^{n x_1} + e^{n x_2} + \dots + e^{n x_n}),$$

has limit.

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0.1.16 Let $(G, *)$, $G \subseteq \mathbb{R}$ be a group such that there exist a continuous isomorphism $f : (G, *) \rightarrow (\mathbb{R}_+^*, \cdot)$. Study the sequence $(x_n)_{n \geq 0}$,

$$x_{n+2} = x_n * x_{n+1}, \quad n \in \mathbb{N}, \quad x_0, x_1 \in G,$$

for convergence.

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0.1.17 Prove the following equality

$$\sum_{n=1}^{\infty} \arctan \frac{1}{n^2} = \arctan \frac{\tan \frac{\pi}{\sqrt{2}} - \operatorname{th} \frac{\pi}{\sqrt{2}}}{\tan \frac{\pi}{\sqrt{2}} + \operatorname{th} \frac{\pi}{\sqrt{2}}}.$$

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0.2 Exercises: Series of Numbers

Determine whether each of the following series is convergent or not.

0.2.1 $\sum_{n=1}^{\infty} \frac{(2n)!!}{n^n}.$

0.2.2 $\sum_{n=1}^{\infty} a^n \left(1 + \frac{1}{n}\right)^{n^2+bn+c},$

$a, b, c \in \mathbb{R}, a \geq 0.$

0.2.3 $\sum_{n=1}^{\infty} a^{n^b},$

$a, b > 0.$

0.2.4 $\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}.$

0.2.5 $\sum_{n=1}^{\infty} \frac{n!}{(a+1)(a+2)\cdots(a+n)},$

$a > -1.$

0.2.6 $\sum_{n \geq 1} n \left(\left(1 + \frac{1}{n+1}\right)^{n+1} - \left(1 + \frac{1}{n}\right)^n \right).$

0.2.7 $\sum_{n \geq 2} n^{-1-1/\sqrt{\log_2 n}}.$

Find the sum of the following series:

0.2.8 $\sum_{n=1}^{\infty} \frac{1}{(a+n)(a+n+1)\cdots(a+n+p)},$

$a > -1, p \in \mathbb{N}^*.$

0.2.9 $\sum_{n=1}^{\infty} \left(\sum_{k=1}^n k(k+1)\cdots(k+p) \right)^{-1},$

$p \in \mathbb{N}^*.$

0.2.10 $\sum_{n=1}^{\infty} a^{3^n} \frac{a^{3^n} - 1}{a^{3^{n+1}} + 1},$

$a \in \mathbb{R}, a \neq -1.$

0.2.11 $\sum_{n=0}^{\infty} \arctan \frac{1}{n^2 + n + 1}.$

0.2.12 $\sum_{n=0}^{\infty} \arctan \frac{1}{2n^2}.$

0.2.13 $\sum_{n=0}^{\infty} \arctan \frac{2}{n^2}.$

0.2.14 $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$

0.2.15 $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)}.$

0.2.16 $\sum_{n=0}^{\infty} \left\lfloor \frac{x + 2^n}{2^{n+1}} \right\rfloor,$

$x \in \mathbb{R}.$

$$0.2.17 \quad \sum_{n \geq 0} \frac{(2n)!!}{(2n+1)!!} a^n, \quad a \in [-1, 1).$$

$$0.2.18 \quad \sum_{n \geq 0} \frac{(2n-1)!!}{(2n)!!} a^n, \quad a \in [-1, 1).$$

$$0.2.19 \quad \sum_{n \geq 0} \frac{(n!)^2}{(2n+1)!} a^n, \quad a \in [-4, 4).$$

$$0.2.20 \quad \sum_{n \geq 0} \frac{1}{\binom{2n}{n}}.$$

0.2.21 Consider the sequence $(S_p)_{p \geq 1}$,

$$S_p = \sum_{n \geq 1} \frac{n^p}{n!}, \quad p = 1, 2, \dots$$

Find a recurrence formula for S_p and the terms S_1, S_2 and S_3 .

Use the notions presented in the course, to prove the following:

0.2.22

$$\sum_{n=1}^{\infty} \arctan \frac{1}{n^2} = \arctan \frac{\tan \frac{\pi}{\sqrt{2}} - \operatorname{th} \frac{\pi}{\sqrt{2}}}{\tan \frac{\pi}{\sqrt{2}} + \operatorname{th} \frac{\pi}{\sqrt{2}}}.$$